Stephanie Wang

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Brief intro

I have worked and published papers in the fields of **geometry processing** (explicit / implicit representations, parameterization, mesh processing, geometric optimization, geometric deep learning) and **physics-based simulation** of all kinds of materials (solids, fluids, waves, cloth, hair, friction / contact, deformable bodies, plasticity, visco-elasto-plasticity, and fracture / damage). I am looking for **academic positions** where I can continue to explore mathematical models and solutions to problems arising in computer graphics and help PhD students grow.

Education

Ph.D. and M.S. in Mathematics, *UCLA*, *Eugene V. Cota-Robles Fellow*. **2014-2020**
Committee: Jeffrey D. Eldredge, Wotao Yin, Luminita Aura Vese, and Joseph M. Teran (advisor)

B.S. in Mathematics, *National Taiwan University*, *magna cum laude*. **2009-2013**

Experience

Research

Research Engineer – reporting to Dr. Qingnan Zhou, *Adobe Inc.*, Seattle, WA. **2024-2025**
Geometry and physics research and software development, with a focus on geometry processing in machine learning and 3D modeling. Mentored students / interns: Yutao Zhang, Baptiste Genest, Dylan Rowe.

Postdoc – with Prof. Albert Chern, *UCSD*, San Diego, CA. **2020-2023**
Geometry processing and physical simulation using mathematical insights from geometric measure theory, exterior calculus, partial differential equations, and optimization theory. Mentored students: Mohammad Sina Nabizadeh, Shiyang Jia, Chad McKell, Hang Yin, Baichuan Wu.
(Note: I took a full-time medical leave between Feb-Aug 2022.)

Ph.D. Study – with Prof. Wilfrid Gangbo, *UCLA*, Los Angeles, CA. **2019-2020**
Regularity theory for minimizers of polyconvex functionals related to Navier-Stokes equation.

Exchange Study – with Prof. Johan Gaume, *EPFL*, Lausanne, Switzerland. **2019 summer**
Simulations and data analysis of snow and tire interaction, avalanche release, and snow micro-structure.

Ph.D Study – with Prof. Joseph Teran, *UCLA*, Los Angeles, CA. **2016-2019**
Physics-based simulations of various materials with Material Point Method and Finite Element Method, using continuum mechanics, convex and nonconvex optimization technique, numerical analysis, parallel computing.

Tech Intern, *Walt Disney Animation Studio*, Burbank, CA. **2018 summer**
R&D for pioneering simulation technology in animated feature films, teaming with VFX artists.

Research Assistant – with Prof. Wen-Wei Lin, *NCTU*, Hsinchu, Taiwan. **2013-2014**
Generalized eigenvalue problems using MATLAB programming.

Teaching

Assistant Adjunct Professor, *UCLA Math Dept*, Los Angeles, CA (virtual). **2020**
Taught remote classes for upper and lower division undergratuate courses: Machine Learning (Math156) and Calculus of Several Variables (Math32A).

Graduate Student Instructor, *UCLA Math Dept*, Los Angeles, CA. **2019 spring**
Taught course: Linear Algebra and Applications (Math33A).

Teaching Assistant, *UCLA Math Dept*, Los Angeles, CA. **2015-2020**
Led discussion sessions and graded homework/exams for 11 undergraduate and graduate level courses: linear algebra and introduction to mathematical proofs (Math 115A), undergrad- and grad-level numerical methods (Math 151B, 269A), introductory, intermediate, and advanced C++ programming (PIC 10A, 10B, 10C).

Awards

Rising Stars in Computer Graphics Research, *WiGRAPH*. **May 2022**

Best Paper Award, *SCA 2019*. **Jul 2019**

Eugene V. Cota-Robles Fellowship, *UCLA*. **Sep 2014**

Dean's Award, College of Science, *National Taiwan University*. **Jun 2013**

Bronze Medal in Applied and Computational Mathematics, *S.T. Yau College Student Mathematics Contest*. **Aug 2012**

Publications

Variational Neural Surfacing of 3D Sketches
Yutao Zhang, Stephanie Wang, Mikhail Bessmeltsev
SA Conference Papers '25: Proceedings of the SIGGRAPH Asia 2025 Conference Papers

Implicit UVs: Real-time Semi-global Parameterization of Implicit Surfaces
Baptiste Genest, Pierre Gueth, Jérémy Levallois, Stephanie Wang
Computer Graphics Forum (Eurographics 2025)

Physical Cyclic Animations
Shiyang Jia, Stephanie Wang, Tzu-Mao Li, Albert Chern
Proceedings of the ACM on Computer Graphics and Interactive Techniques (SCA 2023)

Exterior Calculus in Graphics: Course Notes for a SIGGRAPH 2023 Course
Stephanie Wang, Mohammad Sina Nabizadeh, Albert Chern
SIGGRAPH '23: ACM SIGGRAPH 2023 Courses

Fluid Cohomology
Hang Yin, Mohammad Sina Nabizadeh, Baichuan Wu, Stephanie Wang, Albert Chern
ACM Transactions on Graphics (SIGGRAPH 2023)

Covector Fluids
Mohammad Sina Nabizadeh, Stephanie Wang, Ravi Ramamoorthi, Albert Chern
ACM Transactions on Graphics (SIGGRAPH 2022)

DeepCurrents: Learning Implicit Representations of Shapes with Boundaries
David Palmer, Dmitriy Smirnov, Stephanie Wang, Albert Chern, Justin Solomon
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2022)

Role Detection in Bicycle-Sharing Networks Using Multilayer Stochastic Block Models
Jane Carlen, Jaume de Dios Pont, Cassidy Mentus, Shyr-Shea Chang, Stephanie Wang, Mason A. Porter
Network Science, 2022

Computing minimal surfaces with differential forms
Stephanie Wang and Albert Chern
ACM Transactions on Graphics (SIGGRAPH 2021)

Computational micromechanics of porous brittle solids
Lars Blatny, Henning Löwe, Stephanie Wang, Johan Gaume
Computers and Geotechnics, 2021

A Material Point Method for Elastoplasticity with Ductile Fracture and Frictional Contact
Stephanie Wang
UCLA Doctoral Dissertation, 2020

A thermomechanical material point method for baking and cooking
Mengyuan Ding, Xuchen Han, Stephanie Wang, Theodore F. Gast, Joseph M. Teran
ACM Transactions on Graphics (SIGGRAPH Asia 2019)

A Hybrid Material Point Method for Frictional Contact with Diverse Materials
Xuchen Han, Theodore F. Gast, Qi Guo, Stephanie Wang, Chenfanfi Jiang, Joseph M. Teran
Proceedings of the ACM on Computer Graphics and Interactive Techniques (SCA 2019)

Simulation and Visualization of Ductile Fracture with the Material Point Method
Stephanie Wang, Mengyuan Ding, Theodore F. Gast, Leyi Zhu, Steven Gagniere, Chenfanfu Jiang, Joseph M. Teran
Proceedings of the ACM on Computer Graphics and Interactive Techniques (**SCA 2019 Best Paper**)

Preprint

Wave Simulations in Infinite Spacetime
Chad McKell, Mohammad Sina Nabizadeh, Stephanie Wang, Albert Chern

Invited talks

ISTA (Visual Computing Group), Klosterneuburg, Austria.	**Oct 2025**
Polytech Paris-Saclay (GeomeriX), Palaiseau, France.	**Oct 2025**
TU Berlin (Dept. Computer Graphics), Berlin, Germany.	**Oct 2025**
EPFL (Geometric Computing Laboratory), Lausanne, Switzerland.	**Oct 2025**
BU (Prof. Ed Chien Group), Boston, MA.	**Sep 2025**
Adobe Research, (virtual).	**Nov 2024**
ICIAM 2023, Tokyo, Japan.	**Aug 2023**
SIGGRAPH 2023 Course, Los Angeles, CA.	**Aug 2023**
NCSU (Geometry and Topology Seminar), Raleigh, NC (virtual).	**Feb 2022**
MIT (CSAIL), Cambridge, MA.	**Nov 2021**
Autodesk Research, (virtual).	**Nov 2021**
Online Seminar Geometric Analysis, (virtual).	**Nov 2021**
Toronto Geometry Colloquium, Toronto, ON (virtual).	**Oct 2021**
Geometry Workshop in Obergurgl 2021, Obergurgl, Austria.	**Sep 2021**
SIGGRAPH 2021 Technical Paper, (virtual).	**Aug 2021**
UCSD (Dept CSE Vis-Comp), San Diego, CA (virtual).	**Apr 2021**
UCSD (CCoM), San Diego, CA (virtual).	**Jan 2021**
CMU (Prof. Keenan Crane Group), Pittsburgh, PA (virtual).	**Dec 2020**
GAMES Webinar, (virtual).	**May 2020**
College of the Holy Cross, Worcester, MA (virtual).	**Nov 2019**
Inria Grenoble-Rhône-Alpes (Elan), Grenoble, France.	**Sep 2019**
ETH Zürich (Computer Graphics Laboratory), Zürich, Switzerland.	**Aug 2019**
EPFL (Snow and Avalanche Simulation Laboratory), Lausanne, Switzerland.	**Aug 2019**
SCA 2019 Technical Paper, Los Angeles, CA.	**Aug 2019**
UCLA (Math GSO Seminar), Los Angeles, CA.	**Nov 2018**

Services

Conference program committee. **2022-present**
ACM SIGGRAPH 2026 technical papers program committee,
ACM SIGGRAPH 2025 technical papers program committee,
SCA 2024 international program committee,
ACM SIGGRAPH 2023 poster jury, and
Eurographics 2023 short paper program committee.

Journals / conferences tertiary reviewer. **2021-present**
ACM SIGGRAPHs, Eurographics, ICIAM, IEEE TVCG.

Research project mentor, *MIT Summer Geometry Initiative*. **2021**

Departmental graduate student representative, *UCLA*. **2017-2020**

Chief Organizer, *Women in Math, UCLA*. **2016-2018**
Created the Women in Math Mentorship Program (WIMMP), volunteered at and organized on campus academic, social, and community outreach events.

Vice President, *Lambda Club, National Taiwan University*. **2012-2013**
Organized academic and social events and grew the community from three members to 30+.

Skills

Programming: C++ (Eigen, tbb), Python (PyTorch, SciPy), lua, MATLAB (CVX), LaTeX, zsh

Tools: Houdini, Vim, git, gdb, valgrind

Math: Optimization, differential geometry, numerical and theoretical PDEs, scientific computing

Languages: English and Mandarin Chinese (bilingual)

Hobbies: Hiking, cooking, and people watching

Last updated: January 22, 2026.